Correction 2026-10-20 (validation sweep B, verdict 110.1: contradicted). The N83 18.528 rho/IC ratio below compares against rho with an empty table on one L0 target. It is superseded and is not an improvement; precomputed rho wins every N83 matched pair (benchmarks/20261016-n83-precomputed-rho-control). The within-IC K1798 vs K1500 result stands.

N83 dense packed-root primary run

2026-10-13 09:42 UTC · exact public point · Apple M4 Pro shared host

The opt-in dense table preserved K1271′s first witness while allocating 4 GiB of packed slots instead of 8 GiB. The pre-panel K1798 capacity arm completed with 298,468 usable base points, 111,004 target probes and 63.324 ms online. The same-point strong-rho control completed in 1,173.293 ms online. The arithmetic rho/IC ratio for this point is 18.528; controlled CPU speedup is unadmitted on this host.

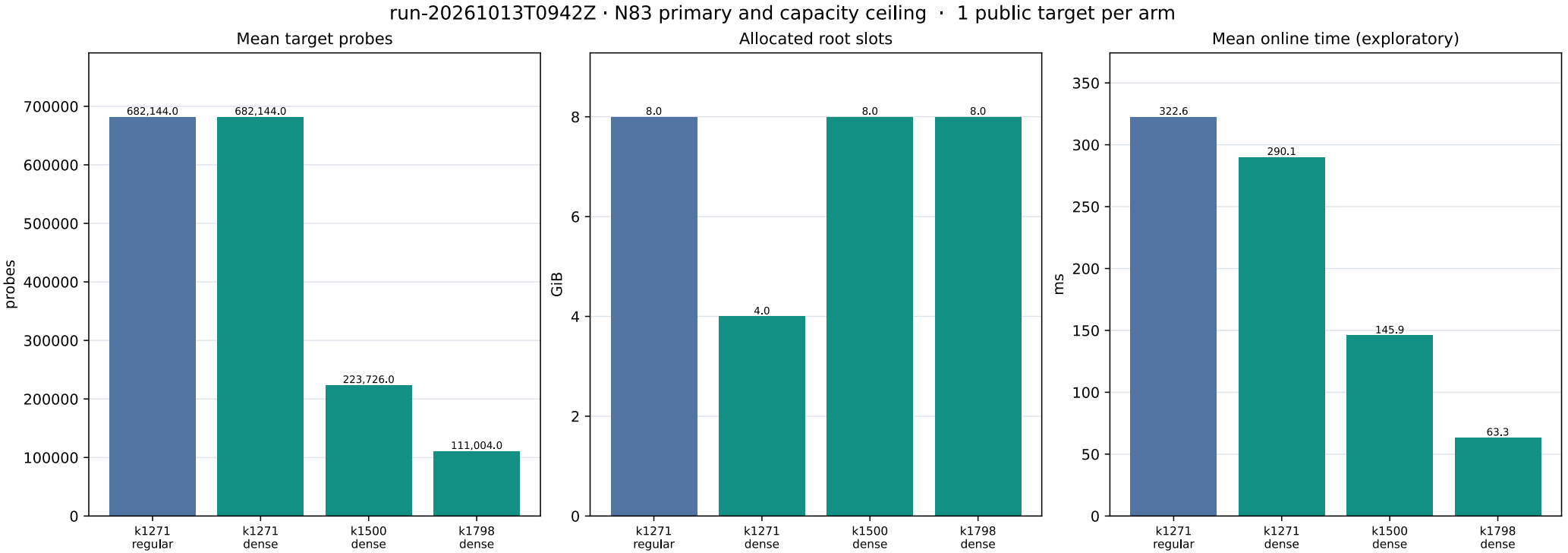


Figure 1: Recorded probes, table allocation and one-target online time. K1500 regular reached the 900 s process cap before a target row; it is retained in the table below.

Arm	Actual B	Columns	Slots	Probes	Online ms	Status
K1271 regular	210,986	1,271	8 GiB	682,144	322.621	replay pass
K1271 dense	210,986	1,271	4 GiB	682,144	290.081	replay pass
K1500 dense	249,000	1,500	8 GiB	223,726	145.943	replay pass
K1500 regular	missing	missing	missing measured	missing	missing	timeout 124
K1798 dense	298,468	1,798	8 GiB	111,004	63.324	replay pass

The exact curve is Koblitz $a = b = 1$ over $F_{2^{83}}$, polynomial low terms $[0, 2, 4, 7]$, subgroup order 8,569,786,107,849,059. All IC arms use rank seed 29, 12 rank threads, one target thread, and the same public point. The online timer starts at target-dependent query work and stops after scalar replay; setup and full-process costs are separate. The selected rho policy is fold-L64-DP6-E0 with seed 2026101301 and one OS thread. Its timer includes point validation, walk/collision and generic scalar replay.

Independent generic-curve replay verified the scalar in all four completed IC arms and the rho receipt. The K1500 regular run used 901.30 s wall including timeout handling and reached 12,855,771,136 B maximum RSS. Another research process was active; no host isolation receipt exists. The 64-point panel was reserved before this run and tests whether the probe ordering persists.

Raw records: `../../runs/primary/`, `../../runs/rho_primary/`, `../../analysis_primary.json`. Source and protocol: `../../PROTOCOL.md`, `../../CAPACITY_CEILING.md`; Lean proof map: `../../PROOF_MAP.md`. Producer SHA-256: `caa723d4e2894b8573c04f6ec2ec474ce4fb41eb8a2c452b793b3d2d286aa351`.